

Taeyun Yoo

M.S. in Aviation, Embry-Riddle Aeronautical University, Daytona Beach, FL

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| [Website](#)

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Education

Embry-Riddle Aeronautical University

M.S. in Aviation

Specialization in Aviation Safety Systems

Daytona Beach, FL

Aug 2025–Present

GPA 4.0/4.0

Embry-Riddle Aeronautical University

B.S. in Aeronautical Science

Minor in Air Transportation

Daytona Beach, FL

May 2023

GPA 3.85/4.0

Research Interests

Aviation Safety Analytics; Air Transportation Systems; Trajectory Data Analytics; Terminal and Low-Altitude Airspace Operations

Research Experience

Graduate Researcher

Advisor: Dr. João S. D. Garcia

School of Graduate Studies, College of Aviation

Sep 2025–Present

- ADS-B-based decomposition of approach path stability into sustained and oscillatory deviations
- Wind exposure, approach stability, and go-around initiation in approach and landing operations
- Go-around event-based traffic proximity risk assessment in terminal airspace
- Multi-layer evaluation of AAM disaster response and government-operated public use cases
- ML-based aircraft fuel burn prediction using ADS-B and ACARS data

Graduate Researcher

Advisor: Dr. Ryan Wallace

Center for Advanced Air Mobility

Feb 2026–Present

- LAANC/UASFM-based assessment of shared-airspace exposure between conventional aircraft and UAS operating altitudes
- Nationwide low-altitude traffic density modeling
- ADS-B-based airspace analysis and emerging aviation operations

Publications & Presentations

Publications & Proceedings

[1] Yoo, T., & Garcia, J. S. D. (2026). Data-Driven Framework for Assessing Wind Impacts on Approach Stability. *AIAA AVIATION 2026 Forum*. American Institute of Aeronautics and Astronautics.

Research Presentations

INFORMS Annual Meeting (Oral) – Under Review

Institute for Operations Research and the Management Sciences

From Wind Exposure to Go-Arounds: The Mediating Role of Approach Stability in Commercial Aviation

14th OpenSky Symposium (Oral) – Under Review

OpenSky Network

Beyond Gate-Based Stabilized Approach Criteria: Decomposing Approach Path Stability Using ADS-B Trajectory Data

November 2026

San Francisco, CA

October 2026

Madrid, Spain

UAA 78th Collegiate Aviation Education Conference (Oral) – Accepted	September 2026
University Aviation Association	Denton, TX
<i>The Effect of Wind Exposure on Go-Around Initiation and the Mediating Role of Approach Stability in Flight Operations</i>	
AIAA AVIATION 2026 Forum (Oral)	June 2026
American Institute of Aeronautics and Astronautics	San Diego, CA
<i>Data-Driven Framework for Assessing Wind Impacts on Approach Stability</i>	
Discovery Day (Poster)	April 2026
Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Characterizing Air Traffic Density Using Nationwide ADS-B Data</i>	
COA Fall Research Colloquium (Oral)	November 2025
College of Aviation, Embry-Riddle Aeronautical University	Daytona Beach, FL
<i>Approach Stability Assessment Using ADS-B Data in Flight Training</i>	

Funded Research Projects

2026 Safety Research Seed Grant Program	Boeing Center for Aviation and Aerospace Safety
Graduate Research Lead PI: João S. D. Garcia \$15,000	Jul 2026–Jun 2027
<i>An Empirical Analysis of Go-Around and Collision Risk in Support of In-Time Aviation Safety Management Systems</i>	
Research Activities Supporting AAM National Strategies	Florida Department of Transportation
Graduate Research Assistant PI: João S. D. Garcia \$594,373	Feb–Jun 2026
<i>Advanced Air Mobility in Disaster Response and Other Functions of Government as a Driver for Public Acceptance</i>	

Teaching Experience

Graduate Teaching Assistant	Embry-Riddle Aeronautical University
Supervisor: Dr. Dahai Liu	Dec 2025–Present
Teaching support for:	
• Statistical Analysis for Aviation/Aerospace • Research Methods in Aviation/Aerospace • Graduate Capstone Research Project/Thesis	
Flight Instructor	Aviators Academy
Instructional areas include:	Nov 2023–Jul 2025
• Private Pilot Certificate • Instrument Rating • Commercial Pilot Certificate • Certified Flight Instructor • Certified Flight Instructor-Instrument • Flight Review and Instrument Proficiency Check • Complex Endorsement	

Honors & Awards

Best Poster Award, Embry-Riddle Aeronautical University	April 2026
1st Place, College of Aviation Colloquium, Embry-Riddle Aeronautical University	November 2025
Magna Cum Laude, Embry-Riddle Aeronautical University	May 2023
Dean's List (6 semesters), B.S. in Aeronautical Science, Embry-Riddle Aeronautical University	

Certifications & Licenses

Commercial Pilot – ASEL, AMEL, Instrument Airplane
Flight Instructor – ASEL, AMEL, Instrument Airplane
Ground Instructor – Advanced, Instrument
Remote Pilot – sUAS

Technical Skills

Python, SPSS, LaTeX

Professional Memberships

Institute for Operations Research and the Management Sciences (INFORMS)
American Institute of Aeronautics and Astronautics (AIAA)
University Aviation Association (UAA)
Royal Aeronautical Society (RAeS)

Leadership & Service

Secretary

Korean Student Association

Embry-Riddle Aeronautical University

2019–2020

Administrative Specialist; Squad Commander

Republic of Korea Military Service

Korea Defense Communication Command

2017–2019

Vice Captain

Table Tennis Club

University of North Dakota

2016–2017